Step 5A: Sparse PLS Comparison

Sparse PLS keeps the PLS idea but forces each component to use only a limited number of variables. This implementation is a transparent sparse-PLS approximation: within each component, it keeps only the largest predictor weights before deflation. This is useful here because ordinary PLS can spread importance across correlated variables, while sparse PLS asks which variables still matter when each latent component must be simpler.

Model Comparison

Model	Predictors	Components	CV R2	CV RMSE
Full unfiltered PLS	70	12	0.565	0.057
Step 3 cleaned PLS	27	5	0.558	0.058
Sparse PLS	27	5	0.560	0.058

Best sparse setting: keep 15 variables per component.

Top Sparse PLS Variables

Variable	Family	VIP	In Sparse Weights	Components
block3_visible_minority_share	immigration_citizenship	2.217	True	3
block1_bachelors_or_higher_25_64_share	household_class	1.921	True	2
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.861	True	3
block1_average_household_size	household_class	1.731	True	3
block3_non_citizen_share	immigration_citizenship	1.444	True	3
block4_mayoral_top_two_margin	mayoral_competitiveness	1.211	True	2
block3_recent_immigrant_share	immigration_citizenship	1.146	True	5
block4_federal_margin	federal_competitiveness	0.956	True	3
block1_age_65_plus_share	age_structure	0.877	True	3
block5_school_age_5_17_share	service_contact	0.823	True	4
block1_age_18_34_share	age_structure	0.753	True	2
block5_no_car_household_share	transportation_access	0.737	True	5
block1_low_income_lim_at_share	household_class	0.721	True	3
block4_provincial_margin	provincial_competitiveness	0.719	True	3
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.711	True	3
block5_social_housing_share	service_contact	0.689	True	2
block2_apartment_share	housing_form_density	0.583	True	3
block2_condo_share	housing_form_density	0.540	True	3
block5_shelter_access_1200m	service_access	0.529	True	2
block2_population_density_per_km2	housing_form_density	0.449	True	2

Interpretation

Sparse PLS is mainly a readability test. If the same variables remain important under sparsity, that strengthens the story. If performance collapses, it suggests turnout is being explained by a wider bundled social geography rather than a small handful of variables.

Sparse PLS did not introduce new raw variables beyond the cleaned predictor set. Instead, it reweighted the same cleaned variables under a sparsity constraint. The variables that remain strongest under this stricter setup are visible minority share, bachelor or higher education share, effective mayoral candidates above five percent, average household size, non citizen share, mayoral top two margin, recent immigrant share, federal margin, age 65 plus share, school age children share, age 18 to 34 share, no car household share. This supports the same reference categories as the cleaned PLS model: immigration and racialized geography, education and class resources, household composition, electoral competitiveness, age structure, and selected service-contact context.